Atmosic SDK

User Guide

SUMMARY: This document provides instructions on how to install the Atmosic Software Development Kit (SDK) and use it with the Atmosic Evaluation Kit (EVK) based on ATM2/ATM3 and ATM33/e Wireless SoC Devices.

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Atmosic™

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Table of Contents

Acronyms and Abbreviations 5

1. Overview 6

2. EVK Setup 6

2.1 ATM2/ATM3 EVK 6

2.2 ATM33/e EVK 7

3. SDK Installation 9

3.1 Using Windows Installer 9

3.2 Install Atmosic SDK 9

3.3 Install Atmosic Reference Design Interface (RDI) 13

3.3.1 ATM2/ATM3 EVK 13

3.3.2 ATM33/e EVK 15

3.3.3 Remove RDI Interface 18

3.4 Connect to Debug Interface 18

3.5 Launch MSYS2 shell 19

3.6 Atmosic SDK Platform Working Directory 20

3.7 Linux and macOS Installation 21

4. Platform Tools & Utilities 22

4.1 In-System Programming Tool 22

4.2 build_archive 23

4.3 Makefile Helpers 23

4.4 show_archive 25

4.5 burn_archive 25

4.6 ATM2/ATM3 Platform ROM and OTP Application 27

4.6.1 ROM and Custom Application 27

4.6.2 ROM and Flash Application 28

4.6.3 Flash Operating System 28

5. Make Options and Commands 30

6. Building Beacon Application Using Atmosic SDK 31

6.1 ATM2 and ATM3 Device 31

6.1.1 Board Configuration 31

6.1.2 Beacon 32

6.2 ATM33/e Device 34

6.2.1 Board Configuration 34

6.2.2 Beacon 34

7. Building Energy Harvesting Beacon 35

CONFIDENTIAL

7.1 ATM3 Devices 35
7.2 ATM33e Devices 35

8. Uninstall Atmosic SDK 35

Reference Documents 37

Revision History 38

List of Figures

Figure 1 - ATM33/e Evaluation Kit

Figure 2 - ATM33/e Evaluation Board

Figure 3 - ATM33/e Evaluation Board with 2.4 GHz Whip Antenna

Figure 4 - Extracted Release Package

Figure 5 - Atmosic License Agreement

Figure 6 - SDK Setup Directory

Figure 7 - SDK Installation Path Message

Figure 8 - Installing SDK

Figure 9 - Installation Completed

Figure 10 - SDK in Start Menu

Figure 11 - USB COM Port with ATM2/ATM3 EVK

Figure 12 - Install RDI USB Driver with ATM2/ATM3 EVK

Figure 13 - RDI USB in Device Manager with ATM2/ATM3 EVK

Figure 14 - J-Link Notifications with ATM33/e EVK

Figure 15 - USB COM Port with ATM33/e EVK

Figure 16 - Install RDI USB Driver with ATM33/e EVK

Figure 17 - RDI USB in Device Manager with ATM33/e EVK

Figure 18 - Remove RDI Interface

Figure 19 - Debug Interface Using PuTTY

Figure 20 - Start Menu

Figure 21 - Atmosic SDK Path

Figure 22 - Atmosic Product Code on the EVB

Figure 23 - Uninstall SDK

CONFIDENTIAL

List of Tables

Table 1 - Applicable ATM2/ATM3 EVKs

Table 2 - Applicable ATM33/e EVKs

Table 3 - Binary File Types

Table 4 - build_archive Bundles for ATM2/ATM3 and ATM33 Platforms

CONFIDENTIAL

Acronyms and Abbreviations

Acronyms	Definition
ATM2	ATM2201 ATM2202 ATM2221 ATM2231 ATM2251
ATM3	ATM3201 ATM3202 ATM3221 ATM3231
ATM33	ATM3330 ATM3325
ATM33e	ATM3330e
ATM33/e	ATM33/ATM33e
EVB	Evaluation Board
EVK	Evaluation Kit
GADC	General purpose Analog to Digital Conversion
GPIO	General Purpose Input Output
I2C	Inter-Integrated Circuit
IDE	Integrated Development Environment
NVDS	Non-Volatile Data Storage
OB	On Board
OTP	One Time Programmable
RDI	Remote Debug Interface
SDK	Software Development Kit
SoC	System on Chip

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1. Overview

This document provides instructions on how to install the Atmosic SDK to support EVK based on the ATM2/ATM3/ATM33/ATM33e Wireless SoC Devices. For more information about using the EVK, please refer to the EVK User Guides.

For third party IDEs, refer to IAR Workbench User Guide, SEGGER Embedded Studio User Guide, or Keil MDK Configuration Guide for installation and configuration, and IDE Auxiliary Flash Programming Tool User Guide available on the Atmosic Support website. Please submit a support request for access credentials to the Atmosic Support website.

Related documents are listed in the Reference Documents section.

2. EVK Setup

2.1 ATM2/ATM3 EVK

There are multiple versions of the ATM2 or ATM3 EVKs based on the specific device and package configuration. For hardware setup and configuration instructions, consult the User Guide for the specific EVK being used. The latest version of EVK User Guides can be found on the Atmosic Support website.

Table 1 shows the ATM2/ATM3 EVKs supported.

EVK	Kit Part Number
Evaluation Kit for ATM2202	ATMEVK-M2202-02
Evaluation Kit for ATM2221	ATMEVK-M2221-02
Evaluation Kit for ATM2251	ATMEVK-M2251-01
Evaluation Kit for ATM3201	ATMEVK-M3201-02
Evaluation Kit for ATM3202	ATMEVK-M3202-02
Evaluation Kit for ATM3221	ATMEVK-M3221-02

Table 1 - ATM2/ATM3 EVKs Supported

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2.2 ATM33/e EVK

Table 2 shows the applicable ATM33/e EVKs.

EVK	Kit Part Number
Evaluation Kit for ATM3325	ATMEVK-3325-QK
Evaluation Kit for ATM3325 with Extended Storage Package	ATMEVK-3325-LQK
Evaluation Kit for ATM3330	ATMEVK-3330-QN
Evaluation Kit for ATM3330e	ATMEVK-3330e-QN

Table 2 - Applicable ATM33/e EVKs

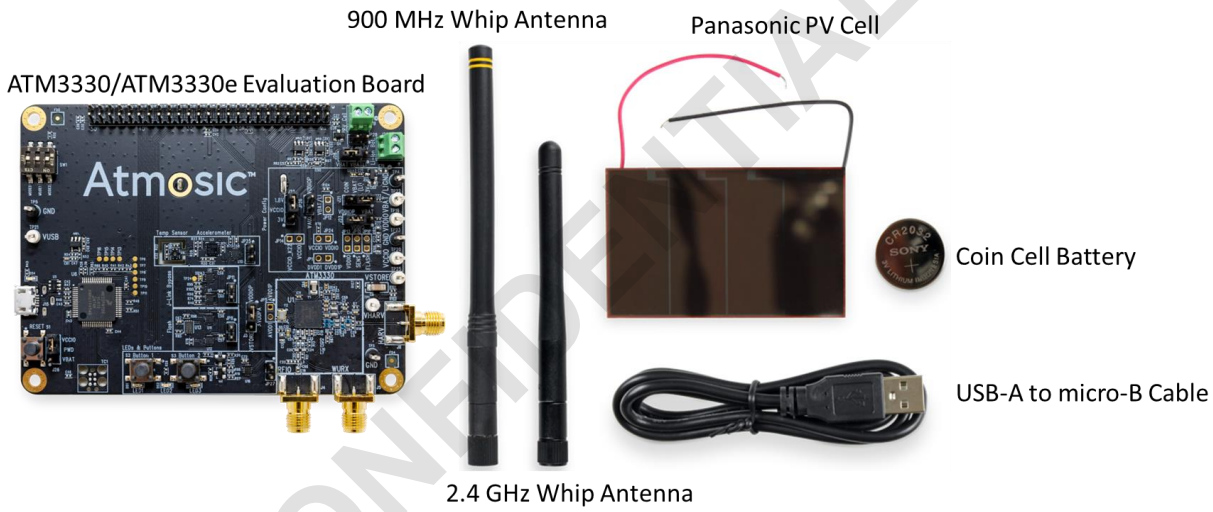


Figure 1 - ATM33/e Evaluation Kit

1) Connect the ATM33/e EVB Micro-B USB connector as shown in Figure 2 to the computer using the Micro-B USB cable provided.

2) Attach the 2.4 GHz whip antenna to the RFIO connector as shown in Figure 3.

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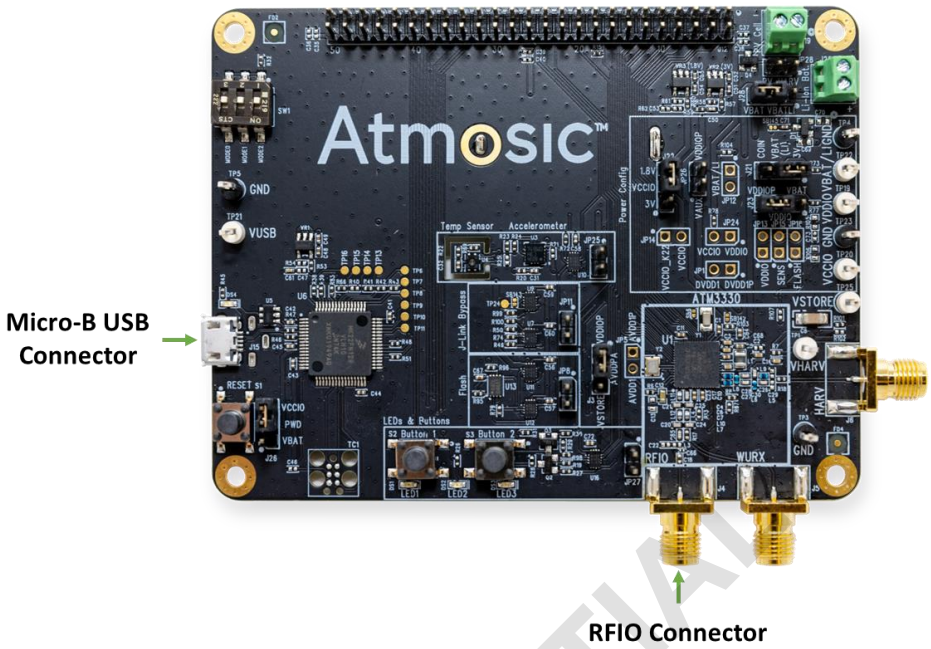


Figure 2 - ATM33/e Evaluation Board



Figure 3 - ATM33/e Evaluation Board with 2.4 GHz Whip Antenna

3) Install COIN battery when VBATT = COIN OR configure EVB to run on VBATT = 3 V. Refer to the **ATM33/e Series Evaluation Kit User Guide** for correct jumper settings.

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3. SDK Installation

3.1 Using Windows Installer

The SDK Windows Installer which supports Windows 10 and later is available on the Atmosic Support website. The installation requires administrator privileges.

3.2 Install Atmosic SDK

The setup package of Windows Installer for Atmosic SDK is a compressed file named AtmosicSDK_Inst_<version_number>.zip. Please extract the file and double click the AtmosicSDK_Inst_<version_number>.exe to start the Atmosic SDK Installer. Figure 4 shows the release package source tree after the files are extracted.

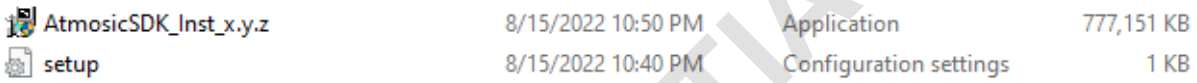


Figure 4 - Extracted Release Package

Please review the license agreement at Atmosic website. If you accept the terms of the agreement, click I Agree to continue.

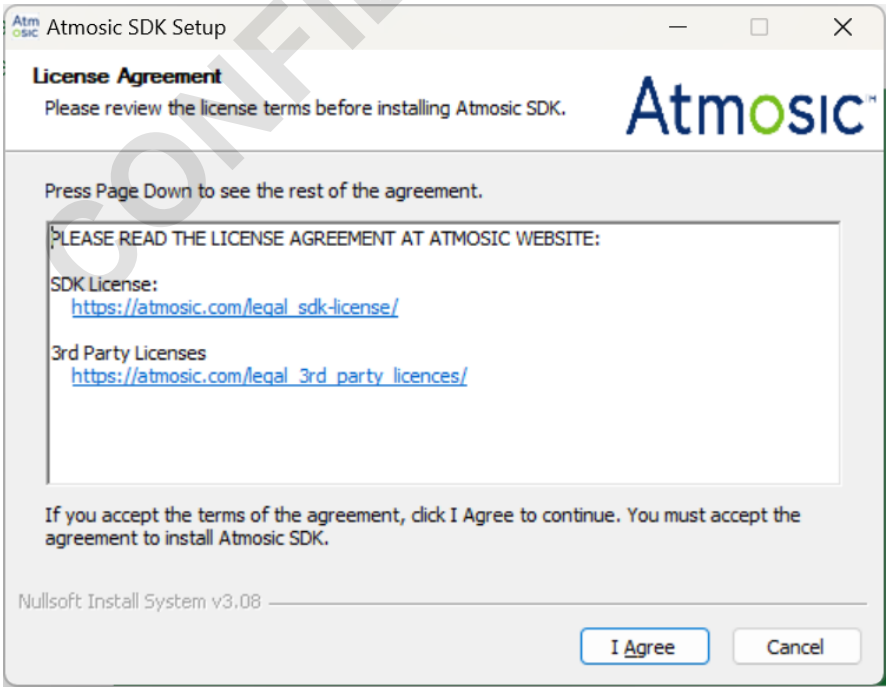


Figure 5 - Atmosic License Agreement

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The Atmosic SDK will occupy 2.5 GB disk space and will be located in the C:\AtmosicSDK\ folder, see Figure 6. Please make sure the disk space is enough before installing the Atmosic SDK. Subsequently, click the Install button to continue.

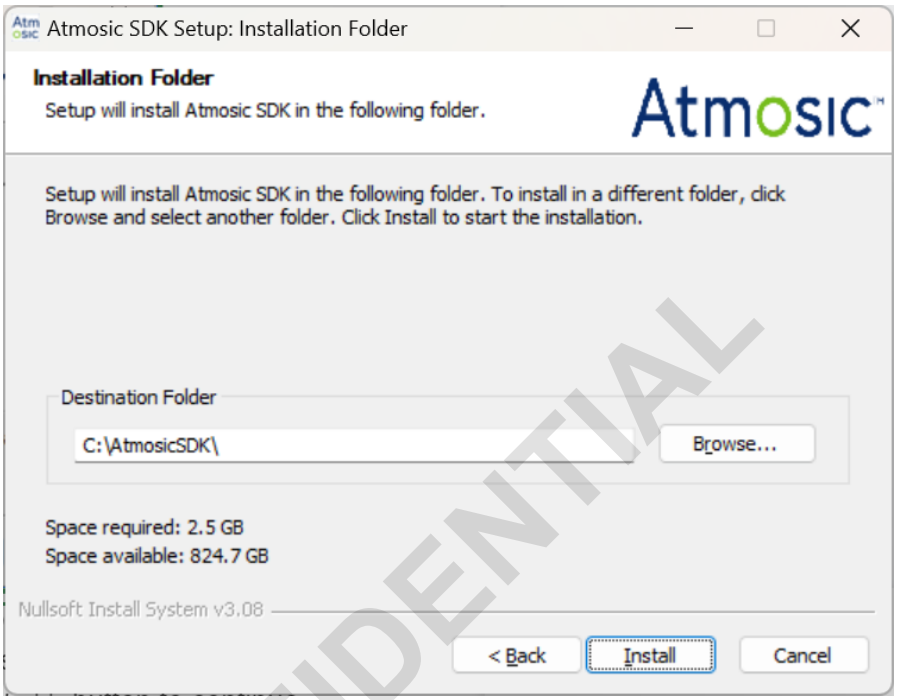


Figure 6 - SDK Installation Folder

***Note**: The installer cannot accept a path destination directory name with spaces. For example, `Atmosic SDK` is not acceptable while `AtmosicSDK` is acceptable. The Windows Installer will pop-up a message if the user selects an installation path name that includes spaces. See Figure 7.*

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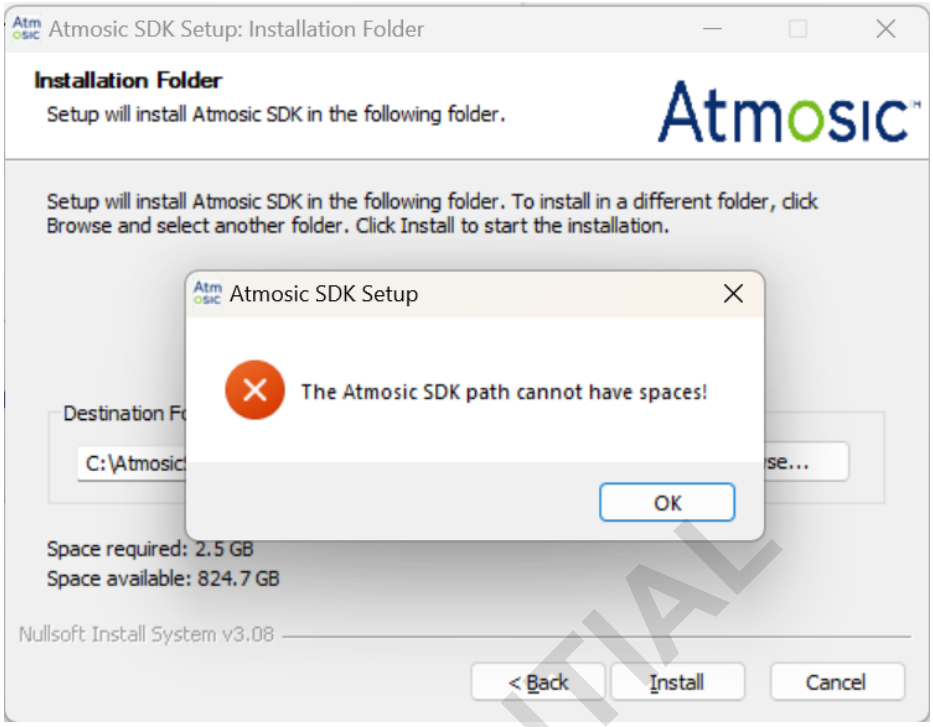


Figure 7 - SDK Installation Path Message

The installation process takes several minutes to complete. The Windows Installer is installing MSYS2, GNU Toolchain, and Atmosic SDK files to the target directory. See Figure 8.

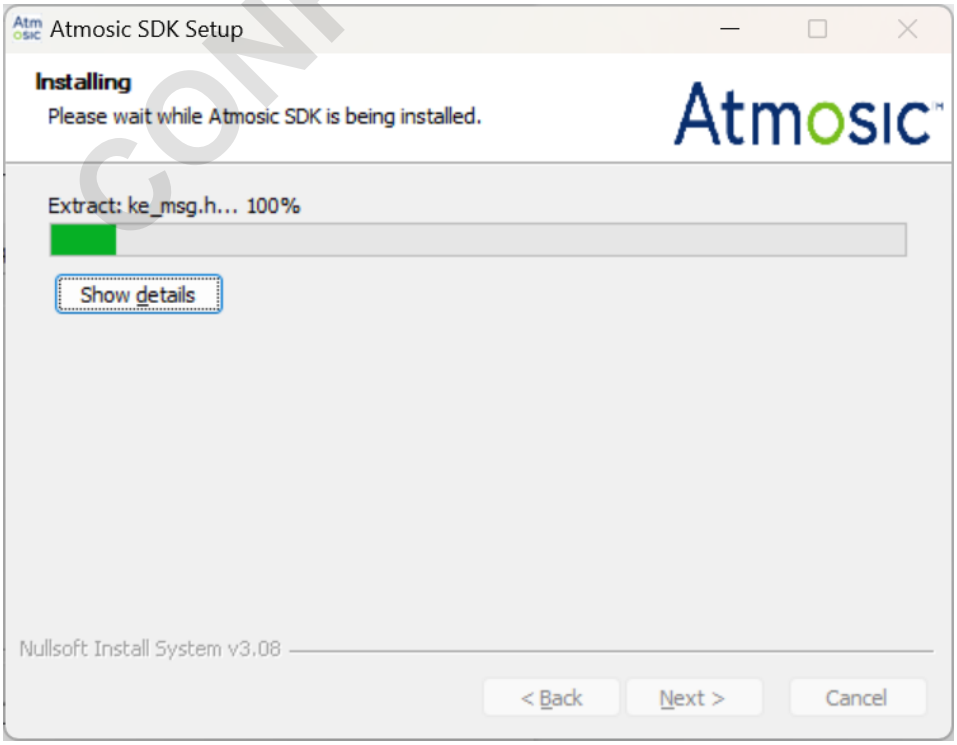


Figure 8 - Installing SDK

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The Windows Installer will pop up a message to notify the installation completed, click OK to close the Windows Installer. See Figure 9.

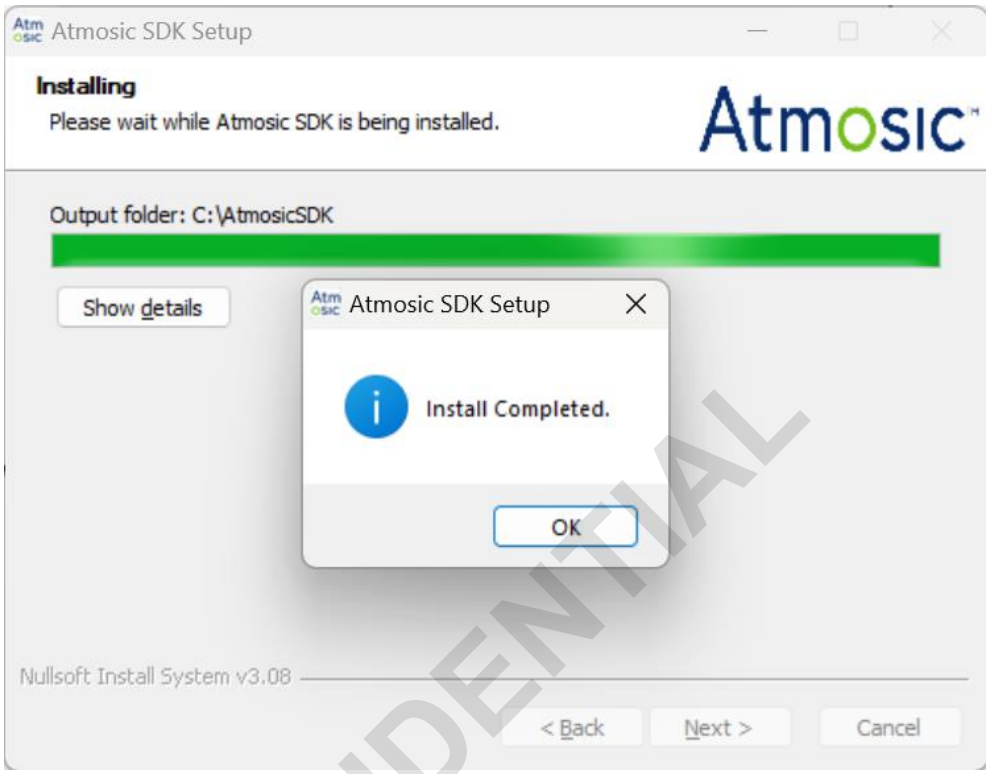


Figure 9 - Installation Completed

The Installer will create a folder named AtmosicSDK in the Start Menu as shown in Figure 10 during the installation process.

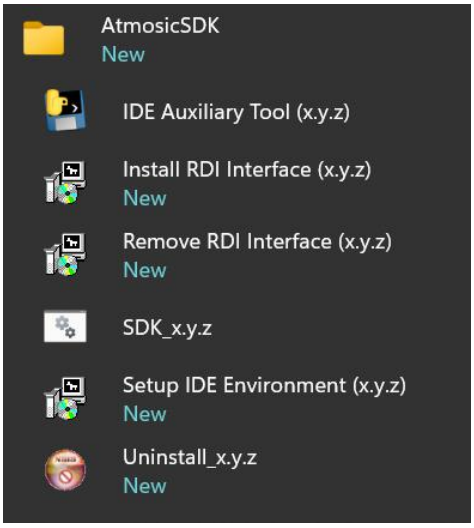


Figure 10 - SDK in Start Menu

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3.3 Install Atmosic Reference Design Interface (RDI)

3.3.1 ATM2/ATM3 EVK

Please connect the Atmosic EVB (USB1) to the Interface board, and connect the USB cable to the laptop and wait for Windows to install the FTDI driver (~ 30 seconds).

Open the device manager (open the Run dialog box by pressing and holding the Windows key, then press the `R` key).

Enter `devmgmt.msc` then click the OK button to verify whether the RDI USB interface exists or not.

If there are 2 COM ports listed in the device manager after plugging-in the Atmosic EVB to the laptop, please click the Install RDI Interface shortcut in `Start Menu/Atmosic` folder. See Figure 11.

***Note**: The COM port numbers are assigned by the Windows system.*

***Note**: If the device manager shows Atmosic RDI USB driver is already installed, move to the Connect To Debug Interface section.*

***Note**: Atmosic RDI USB driver installation is required when the new interface board is connected to the computer for the first time.*

***Note**: Refer to the Reference Documents section for more information regarding the Atmosic EVKs.*

The Device Manager shows COM port information before installing the Atmosic RDI USB driver.

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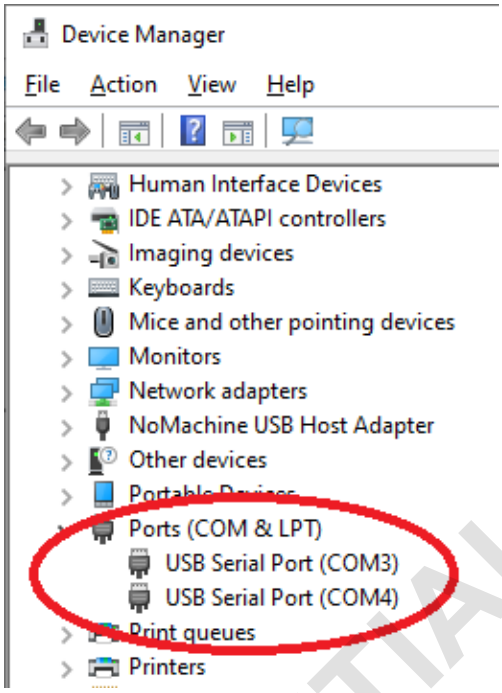


Figure 11 - USB COM Port with ATM2/ATM3 EVK

Please click on the `Install` button if the system pop-up the following message.

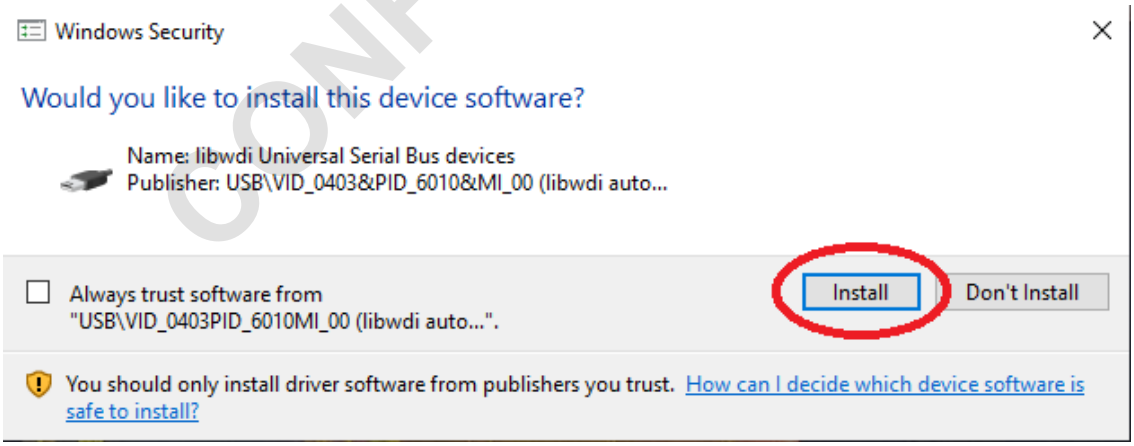


Figure 12 - Install RDI USB Driver with ATM2/ATM3 EVK

After installing Atmosic RDI USB driver, the Atmosic RDI USB1 interface will display in the Device Manager. Moreover, the COM4 is a debug console that could be used to display debug messages through PuTTY.

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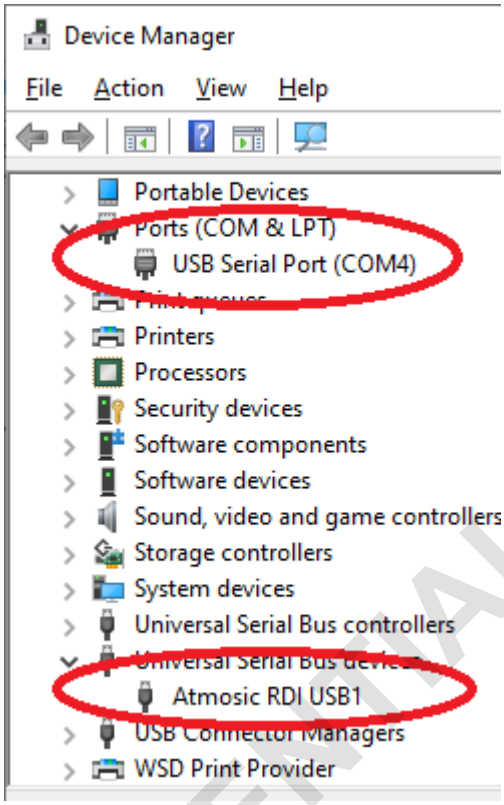


Figure 13 - RDI USB in Device Manager with ATM2/ATM3 EVK

3.3.2 ATM33/e EVK

Please connect the Atmosic EVB to the laptop and wait ~ 30 seconds. First time installation, a J-Link notification will pop-up in Figure 14. Allow it to finish setting up as shown in Figure 14. This will not be displayed in the device manager.

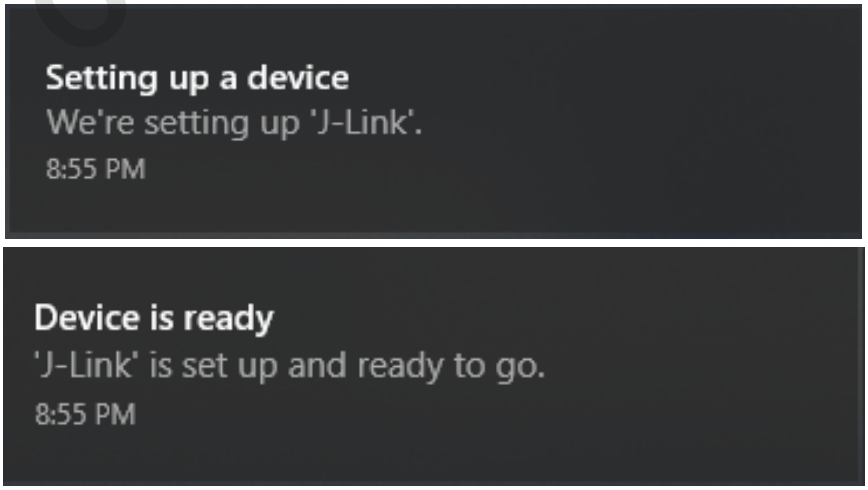


Figure 14 - J-Link Notifications with ATM33/e EVK

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Open the device manager (open the `Run` dialog box by pressing and holding the Windows key, then press the `R` key).

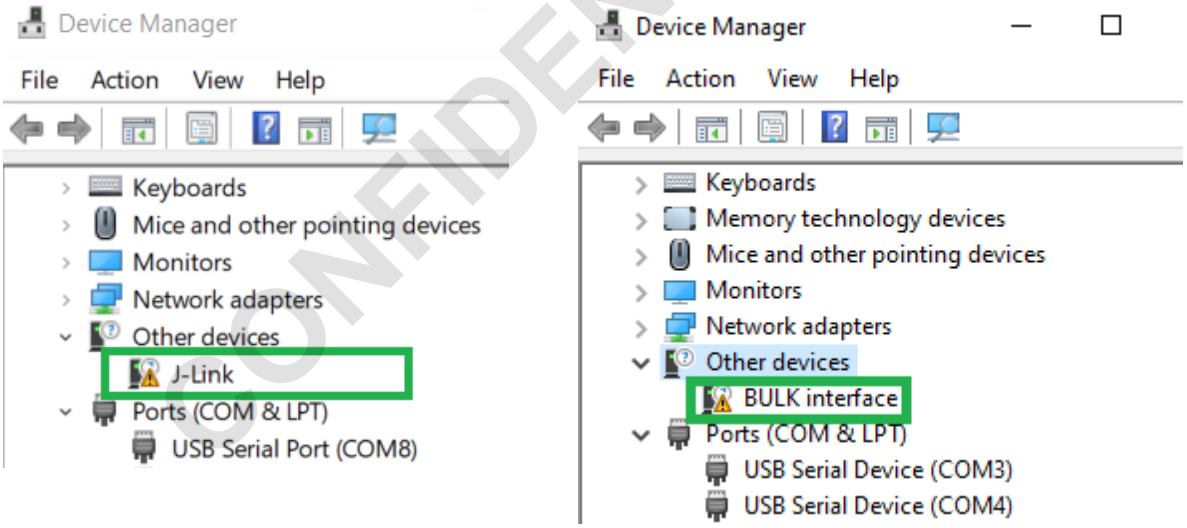
Enter `devmgmt.msc` then click the `OK` button to verify whether the J-Link OB interface or BULK interface exists or not.

If there is a J-Link or BULK interface listed in the device manager under `Other devices`, please click the `Install RDI Interface` shortcut in `Start Menu/Atmosic` folder.

***Note**: The COM port numbers are assigned by the Windows system.*

***Note**: If the device manager shows Install Completed, move to the next section.*

Open Device Manager before installing the RDI Interface driver.



If J-Link driver already installed

No previously installed J-Link driver

Figure 15 - USB COM Port with ATM33/e EVK

Please click the `Install` button if the system pop-up the following message.

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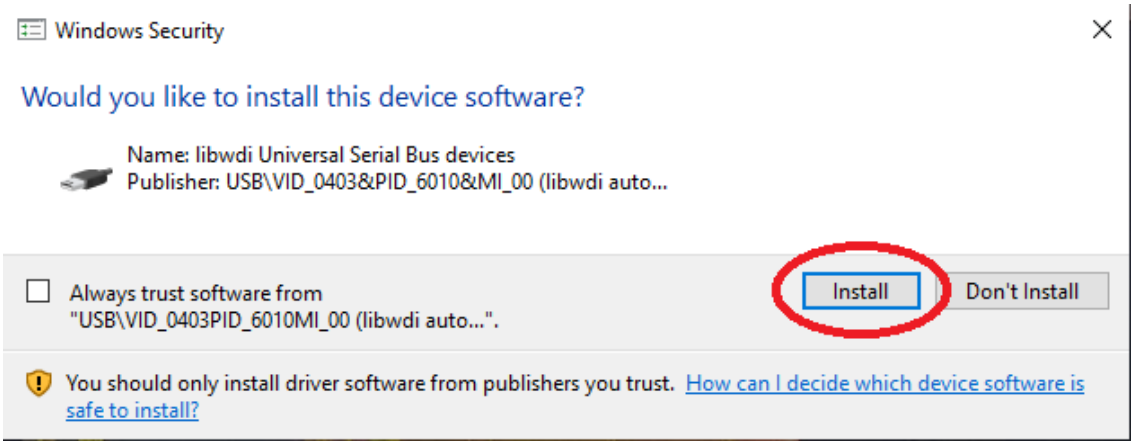


Figure 16 - Install RDI USB Driver with ATM33/e EVK

After installing Atmosic RDI USB driver, verify the successful installation and confirm that the BULK interface shows up under Universal Serial Bus devices in Device Manager. Moreover, one of the serial COM ports shown in these examples will be the debug console output that displays the debug messages through PuTTY.

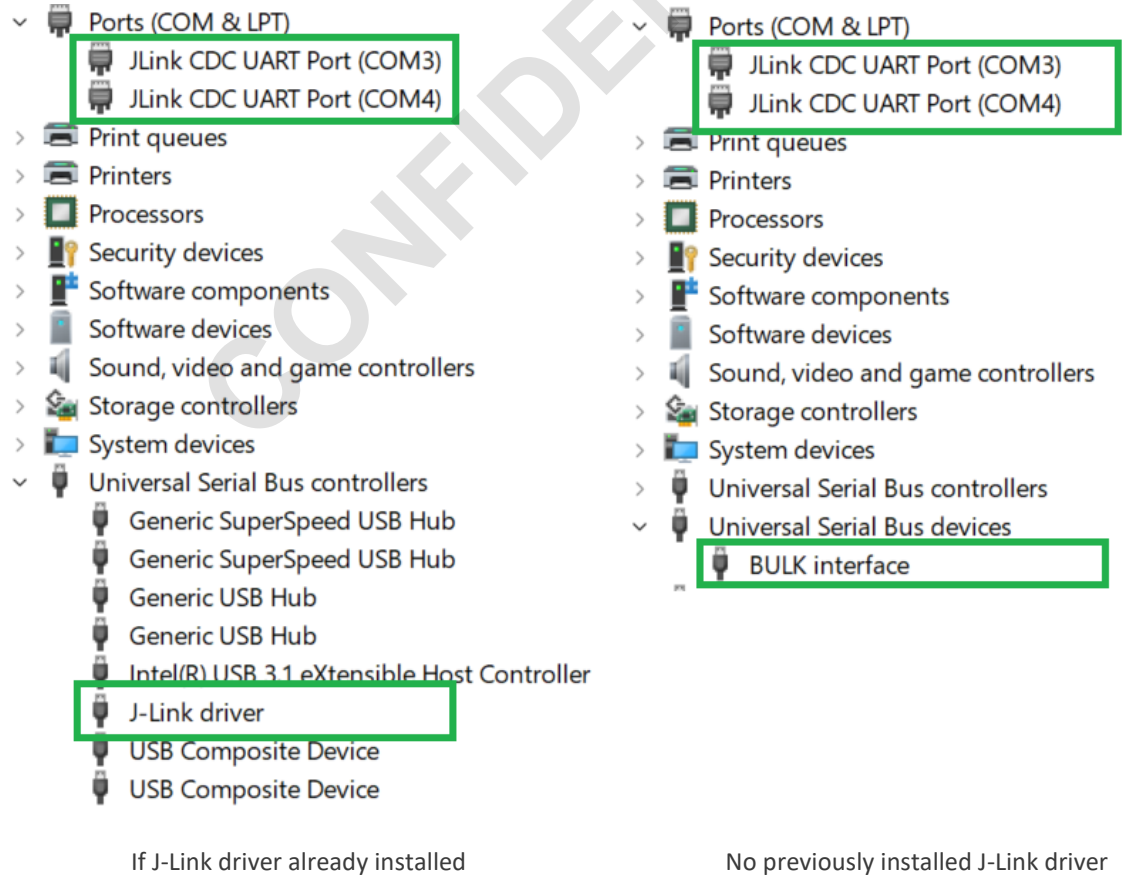


Figure 17 - RDI USB in Device Manager with ATM33/e EVK

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3.3.3 Remove RDI Interface

If you want to remove the RDI driver or switch to JLink, please click the Remove RDI Interface shortcut in Start Menu/Atmosic folder as shown in Figure 18.

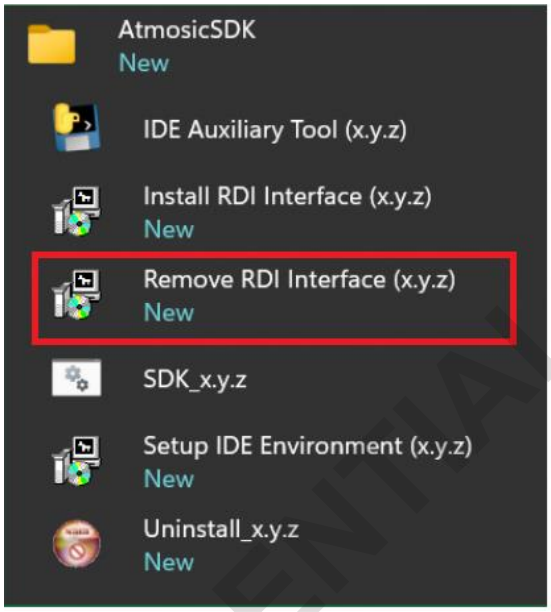


Figure 18 - Remove RDI Interface with ATM33/e EVK

3.4 Connect to Debug Interface

The Atmosic EVK provides the log output through a debug interface (COM port). The developer could use utility to capture (i.e. PuTTY, Tera Term ...etc.) the log. This section describes how to capture logs through PuTTY.

When opening the PuTTY tool, click the

1) `Serial` radio button then input the COM port information in
2) `Serial line` item (i.e. COM11),

and input 115200 in

3) `Speed` item.

Subsequently, click the

4) `Open` button to open the COM port and wait for debug message output.

The debug console will only begin to display when an example is running; otherwise the console will remain empty as shown in Figure 19.

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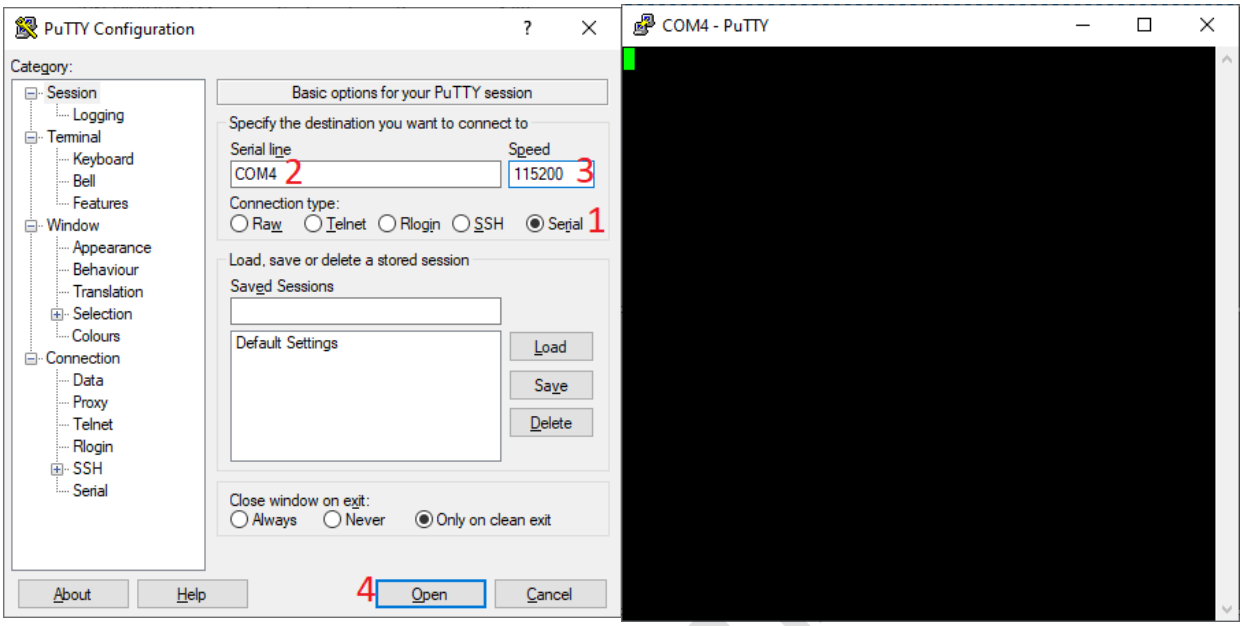


Figure 19 - Debug Interface Using PuTTY

3.5 Launch MSYS2 shell

Invoke the MSYS2 shell from the Windows menu, Start Menu/AtmosicSDK/SDK_<version_number>. Figure 20 shows the Start Menu.

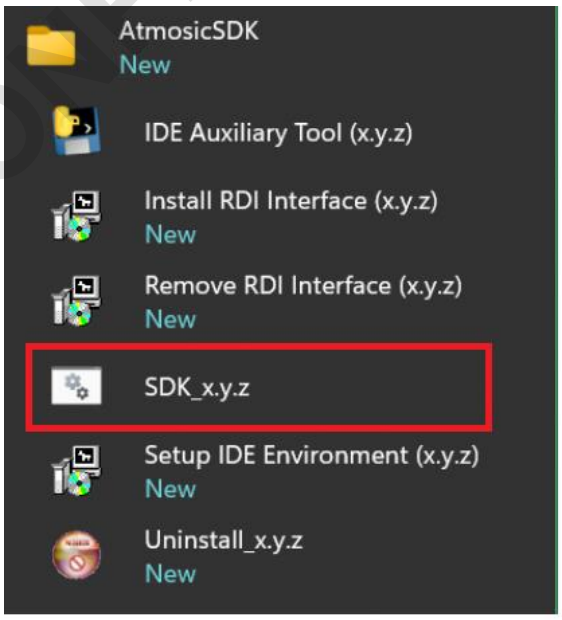


Figure 20 - Start Menu

The MSYS2 shell will be invoked and located at the AtmosicSDK working directory as shown in Figure 21.

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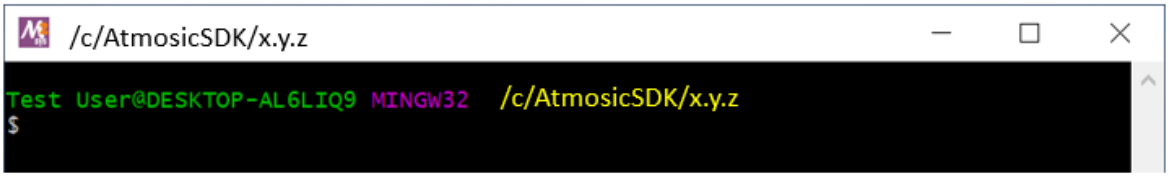


Figure 21 - Atmosic SDK Path

3.6 Atmosic SDK Platform Working Directory

SDK 5.2.0 and later supports ATM2/ATM3 and ATM33/e devices. The difference between ATM2 and ATM3 device versions is captured by the second digit in the -xxx extension of the product code for the device.

In order to build the correct application software for any ATM2, ATM3, or ATM33/e devices, it is necessary to identify the device version and use the appropriate directory in the Atmosic SDK for the software build.

ATM2 and ATM3 EVB have a sticker that identifies the part number of the Atmosic device on the board. For example, the Atmosic EVB is labeled with ATM3221-010 as shown in Figure 22. It is assembled with an ATM32xx-x1x chip revision and to compile for this revision, the platform path is platform/atm3/ATM32xx-x1x. Refer to the Atmosic Chip Revisions Application Note for information regarding ATM2/ATM3 chip revisions.

The ATM3 EVB has printed the part number of the Atmosic device on the board, please refer to Figure 22.

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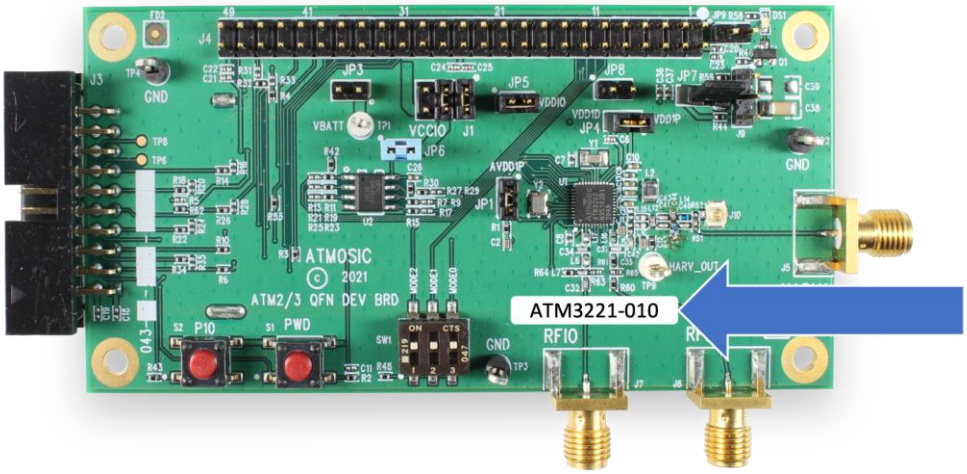


Figure 22 - Atmosic Product Code on the EVB

Following is a list of all Atmosic SDK platform working directories, please use the appropriate directory in the Atmosic SDK for the software build.

- platform/atm2/ATM22xx-x0x
- platform/atm2/ATM22xx-x1x
- platform/atm3/ATM32xx-x0x
- platform/atm3/ATM32xx-x1x
- platform/atm33/ATM33xx-5

3.7 Linux and macOS Installation

For Linux, see SDK_root/tools/README for install instructions.

For macOS, see SDK_root/tools/README.macOS for install instructions.

Ensure BLE_adv example can compile without error after executing the following commands for ATM2/ATM3 platforms:

```
$ cd platform/atm2/ATM22xx-x0x/examples/BLE_adv/
$ make
```

ATM2/ATM3 Make Executed without Error for BLE_adv Sample Application:

```
/c/AtmosicSDK/x.y.z/tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-
objcopy -O binary BLE_adv.elf BLE_adv.bin
/c/AtmosicSDK/x.y.z/tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-
objdump -d BLE_adv.elf >BLE_adv.asm
```

Ensure BLE_adv example can compile without error after executing the following commands for ATM33/e platforms:

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```
$ cd platform/atm33/ATM33xx-5/examples/BLE_adv/
$ make
```

ATM33 Make Executed without Error for BLE_adv Sample Application:

```
/c/AtmosicSDK/x.y.z/tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-
objcopy -O binary BLE_adv.elf BLE_adv.bin
/c/AtmosicSDK/x.y.z/tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-
objdump -d BLE_adv.elf >BLE_adv.asm
```

***Note**: Reminder to install COIN battery when VBATT = COIN OR configure EVB to run on VBATT = 3 V. Refer to the EVK User Guides for specific EVB for correct jumper settings.*

4. Platform Tools & Utilities

4.1 In-System Programming Tool

The SDK ships with a tool called Atmosic In-System Programming Tool (ISP) for bundling three types of binaries, flash, flash NVDS and OTP NVDS into a single binary archive. See Table 3.

Binary Type	Description
.bin	binary file, contains flash or nvds data only.
.elf	elf file, a common standard file format, consists of elf headers and flash data.
.nvm	OTP NVDS file, contains OTP nvds data.

Table 3 - Binary File Types

The ISP tool, which is also shipped as a stand-alone package, can then be used to unpack the components of the archive and download them on a device.

In every platform example, e.g. BLE_adv_scan, the makefiles have the following targets for building, examining, and programming an ISP archive, respectively.

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```
make build_archive // requires appropriate nvds file, see below
make show_archive // display nvds and image details
make burn_archive // program files into storage
make clean_archive // remove generated binary archive
```

4.2 build_archive

build_archive bundles different binary files from ATM2/ATM3 and ATM33, and $(APP).bin where $(APP) is the name of the application directory. See Table 4.

Platform	Non-OTA support	OTA support
ATM2/ATM3	• $(APP).bin/.elf • flash_nvds.bin	• $(APP).bin/.elf • flash_nvds.bin • otp_nvds.nvm (optional)
ATM33	• bootloader.bin • $(APP).bin • flash_nvds.bin	• mcuboot.bin • atmwstk.bin • flash_nvds.bin • signed_$(APP).bin

Table 4 - build_archive Bundles for ATM2/ATM3 and ATM33 Platforms

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In ATM2/ATM3, not all binaries are mandatory, build_archive can build selective binaries. For example the target $(APP).elf can be chosen instead of the $(APP).bin by setting the make variable `ARCH_FLASH_TYPE=elf`.

***Note**: flash_nvds.bin and $(APP).bin apply only to flash based applications and otp_nvds.nvm applies to OTP storage, see OTP_beacon sample application.*

4.3 Makefile Helpers

Makefile helper allows various application configurations.

Key helpers for NVDS:

```
flash_nvds.data := List of NVDS tags required by application
otp_nvds.data: List of NVDS tags for application running in OTP
```

build_archive NVDS requirements:
For flash-based applications that define flash_nvds.data in makefile:

```
make build_flash_nvds  //  creates flash_nvds.bin
```

For OTP based data that define otp_nvds.data in makefile:

```
make pull_otp_nvds  // creates otp_nvds.nvm
make show_pretty_flash_nvds  // shows contents of .nvm file
make rebuild_otp_nvds // creates new otp_nvds.nvm
```

Output of `make build_archive`

```
$(APP)_arch.atm // archived application, FW binary plus NVDS (OTP and/or
flash)
```

Output of `make build_archive ARCH_FLASH_TYPE=elf`

```
$(APP)_arch.atm // archived application, FW elf plus NVDS (OTP and/or flash)
```

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4.4 show_archive

Example output of show_archive from WURX_ADV example application below. It shows platform and image size:

```
$ make show_archive
/c/AtmosicSDK/x.y.z/tools/atm_isp decode -i arch.atm
Platform ATMx2xx-0x0 / x1x
LoadFlashNvds image=(size=260,content=b"NVDS\x06\x06'\x00"...)
LoadFlash image=(size=1731732,content=b'\x7fELF\x01\x01\x01\x00'...)
```

For ATM33 with OTA support:

```
for ATM33 with OTA support:
/c/AtmosicSDK/5.3.0/tools/atm_isp decode -i TPUTP_server_arch.atm
MPR start: 0, size: 0, lock_size: 0
OTA support: 0
Platform ATM33xx-5 (atm33 family)
LoadRram image=(size=19552,content=b'\x00@\x000\xa9\x87\x01\x10'...)
region_start=65536 region_size=20480 address=268435456 extrainfo=BOOTLOADER
LoadRram image=(size=87940,content=b'\x00\x00\x02 \xbd\xf4\x01\x00'...)
region_start=86016 region_size=497664 address=268435456 extrainfo=APP
LoadRram image=(size=4096,content=b'NVDS\x02\x06\x0eA'...) region_start=583680
region_size=4096 address=268435456 extrainfo=NVDS
```

4.5 burn_archive

The burn_archive argument programs the archived file into the target storage.The atm_isp tool is included with the SDK and is called when running the “make burn_archive” command. CLI can be used to program an archive file by directly calling the atm_isp tool from the tools directory (see below).

Also, the GUI based atm_isp tool can be used to program archive files without the tools provided in SDK. See “isp_tool_gui” provided on customer portal.

```
make burn_archive // programs arch.atm into storage
```

Defining the makefile variable BURN_ARCH_VERIFY tells the ISP tool to verify images after loading them on the device.

BURN_ARCH_DEBUG makes the tool more verbose.

If there is OTP data available, setting BURN_ARCH_ERASE_WORKAROUNDS will erase any unlocked workaround tags (0xfc, 0xfd, and 0xfe) from OTP before burning the new contents.

Finally, defining BURN_ARCH_PROGRAM_ONLY prevents reset-hard-on-exit; it can be used to get a behavior similar to `make program_all` as opposed to `make run_all`.

Using atm_isp tool CLI:

```
$cd <SDK>/tools
$ ../atm_isp burn -h
usage: atm_isp burn [-h] [-i ARCHIVE] [-r OPENOCD_PKG_ROOT] [-E] [-e] [-v]
                    [-c] [-t TCL_SCRIPT] [-d DST_DIR] [-p]

optional arguments:
  -h, --help            show this help message and exit
  -i ARCHIVE, --input ARCHIVE
                        Input archive file
  -r OPENOCD_PKG_ROOT, --openocd_pkg_root OPENOCD_PKG_ROOT
                        Path to directory where openocd and its scripts are
                        found
  -E, --openocd_script_only
                        Stop after preparing OpenOCD script
  -e, --erase_workarounds
                        Erase workaround tags in OTP before loading OTP
  -v, --verbose         Verbose mode
  -c, --check_image     Verify OTP/flash image after burning/loading
  -t TCL_SCRIPT, --tcl_script TCL_SCRIPT
                        Path to output Jim Tcl script for use by OpenOCD
                        (generates Jim Tcl script only; delays all
operations
                        post-unpacking of archive to Tcl/OpenOCD); implies
-E
  -d DST_DIR, --dst_dir DST_DIR
                        Use this directory to dump openocd script in;
implies
                        -E
  -p, --program_only    Program the device only (no reset hard on exit)
```

Example:

```
$ ./atm_isp burn -c -v -i <archive.atm>
```

4.6 ATM2/ATM3 Platform ROM and OTP Application

The OTP_beacon directory provides an example of this mode of operation. A number of pre-configured beacon examples are provided in:

platform/atm[2,3]/ATM<part>/examples/OTP_beacon/reference_beacons.mk
One can be selected by setting the REF_BCN makefile variable when building NVDS.

For example:

```
make REF_BCN:=LR_coded rebuild_otp_nvds -> rebuild OTP based NVDS +
application
make push_otp_nvds -> store into OTP
```

4.6.1 ROM and Custom Application

The on-board OTP is relatively small (4 KB), but that can be enough for certain custom solutions that want to avoid using FLASH. A special NVDS tag can be programmed into OTP that the ROM will copy into RAM and execute.

The tmp1075_sensor_adv directory shows how small applications can be rapidly developed using FLASH and then deployed to OTP.

Workflow for debug application and NVDS in flash:

```
make run_all
```

Deployment of optimized application and NVDS to OTP:

```
make pull_otp_nvds
make rebuild_otp_nvds
make push_otp_nvds
```

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4.6.2 ROM and Flash Application

In this mode, the ROM and an application on FLASH cooperate to share the chip resources. This allows extremely compact yet full Bluetooth applications to be developed using APIs in ROM.

BLE_adv, BLE_scan, HCI, HID_keyboard, and HID_mouse all use this mode of operation. See Make Options and Commands section for the list of make commands.

4.6.3 Flash Operating System

In this mode, the ROM acts as a first stage bootloader for a fully formed operating system on the FLASH. The ROM will configure enough modules to discover the FLASH operating system and map it over the top of the ROM at address 0x00000000.

The image on FLASH begins with the ARM stack pointer, reset vector, and other exception vectors. When the ROM hands execution over to this reset vector, the OS is free to use all chip resources.

By default, the above goals build for and program to the flash on the target. The application will access read-only code and data in place from the flash. For certain applications, this can result in undesirable power consumption and performance.

The AVOID_XIP:=1 option can be specified to locate said code and data in RAM instead, and to have it copied from flash at boot time. Be sure to clean the workspace when switching this option:

```
make clean
make AVOID_XIP:=1 run
```

The makefiles, linker scripts, and openocd scripts default to utilize an entire 4 Mb flash device with the last 32 KB reserved for NVDS.

The FLASH_SIZE and NVDS_SIZE makefile variables can be set on the command line or at the top of the example makefile to accommodate different layouts.

Be sure to clean the workspace when switching either of these options. For example, to build an image suitable for OTP partition programming on a 4 Mb device:

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```
make clean
make FLASH_SIZE:=0x40000
```

To avoid the use of flash entirely, the RUN_IN_RAM:=1 option can be specified. This is particularly useful for the HCI example in order to perform qualification tests on flashless designs.

As above, be sure to clean the workspace when switching between flash and RAM workflows:

```
make clean
make RUN_IN_RAM:=1 run
```

To run the system entirely out of flash, FLASHROM=ble-full-all option can be specified. When using this option, the ROM acts only as a boot loader.

Advantages when using FlashROM build:

- Most up-to-date Bluetooth LE stack and HW workarounds.
- BLE stack can be optimized for specific application.
- RAM partitioning not needed (Refer to Table 2: Memory Reservations for Third-Party OS Images in SDK User Reference Manual).

Disadvantages when using FlashROM build:

- Increased application binary size.
- Increased power consumption due to more flash accesses.
- Some RAM is used for timing critical code sections.

```
make clean
make FLASHROM:=ble-full-all run
```

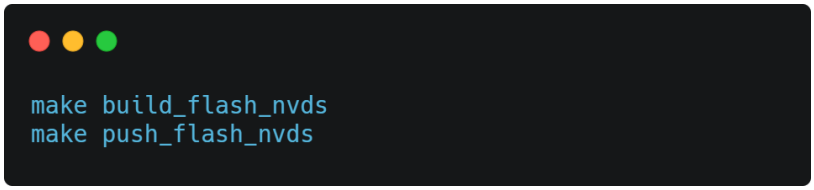
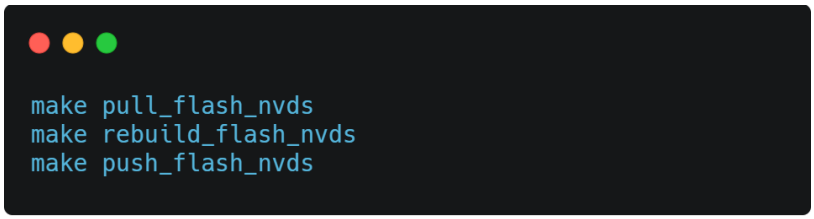
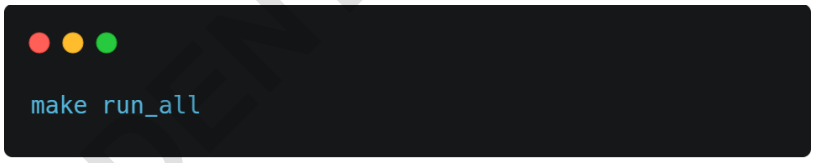
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5. Make Options and Commands

Make Option	Command
Clean build artifacts	`make clean`
Build flash application	`make`
Build flash application, program hardware, but leave ARM M0/M33 CPU halted	`make program`
Build flash application and compare against flash on target	`make verify`
Inspect OTP NVDS from device	`make pull_otp_nvds` `make show_pretty_otp_nvds`
Inspect flash-based NVDS	`make pull_flash_nvds` `make show_pretty_flash_nvds`

Update OTP NVDS on device	`make pull_otp_nvds`
	`make rebuild_otp_nvds`
	`make push_otp_nvds`

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Program new flash-based NVDS	
Update flash-based NVDS	
Build flash NVDS, push, build app, and program	
Build flash NVDS, push, build app, and run	
Collect core dump for offline analysis	
Build archive with protected MP region	

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6. Building Beacon Application Using Atmosic SDK

6.1 ATM2 and ATM3 Device

At MSYS2 shell, go to the BLE_adv example folder to compile the project through the following command.

6.1.1 Board Configuration

The examples that depend on the pinmux driver directly or indirectly (via the

GPIO, I2C, SPI, keyboard, PDM, GADC drivers) will need to specify the board for which the example is being built.

For ATM22xx and ATM32xx devices, the EVB with specific SoC can be specified as: ATM22xx-x0x/ATM32xx-x0x device: m2201, m2202, m2221, m2231, m2251, m3201, m3202, m3221, and m3231.

ATM22xx-x1x/ATM32xx-x1x device: ATMEVK_M2201, ATMEVK_M2202, ATMEVK_M2221, ATMEVK_M2231, ATMEVK_M2251, ATMEVK_M3201, ATMEVK_M3202, ATMEVK_M3221, and ATMEVK_M3231.

6.1.2 Beacon

Beacon configuration: EddyStone, Non Connectable, Scannable, 100 ms Interval.

```
$ cd platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ make BOARD=<chip>
```

If everything is correct, the `make` should execute without error as shown below:

```
user@labtop MINGW32 /c/AtmosicSDK/x.y.z
$ cd platform/atm2/ATM22xx-x1x/examples/BLE_adv

user@labtop MINGW32 /c/AtmosicSDK/x.y.z/platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ make BOARD=m2221
```

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```
user@labtop MINGW32 /c/AtmosicSDK/x.y.z/platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ ls
BLE_adv.asm       atm_gap.o                  pinmux.o
BLE_adv.bin       atm_gap_param.d            pmu.d
BLE_adv.c         atm_gap_param.o            pmu.o
BLE_adv.d         atm_pm.d                   porting_ble.d
BLE_adv.elf       atm_pm.o                   porting_ble.o
BLE_adv.h         ble_gap.d                  reference_beacons.mk
BLE_adv.map       ble_gap.o                  reset.d
BLE_adv.o         ble_module.d               reset.o
README            ble_module.o               rf.d
ROM_errata_10.d   ble_task.d                 rf.o
ROM_errata_10.o   ble_task.o                 rom.elf
ROM_errata_14.d   brwnout.d                  rom.o
ROM_errata_14.o   brwnout.o                  sw_timer.d
ROM_errata_16.d   config_adv_params.h        sw_timer.o
ROM_errata_16.o   ext_flash.d                swd_dbg.d
arm_traceback.d   ext_flash.o                swd_dbg.o
arm_traceback.o   flash_nvds.bin             tag_data
atm_adv.d         hardfault_handler_armv6m.d timer.d
atm_adv.o         hardfault_handler_armv6m.o timer.o
atm_adv_param.d   heap.d                     user_debug.d
atm_adv_param.o   heap.o                     user_debug.o
atm_asm.d         hw_cfg.d                   user_init.d
atm_asm.o         hw_cfg.o                   user_init.o
atm_ble.d         interrupt.d                user_startup_CMSDK_CM0.o
atm_ble.o         interrupt.o                watchdog.d
atm_debug.d       makefile                   watchdog.o
atm_debug.o       partition_info.map
atm_gap.d         pinmux.d
```

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At MSYS2 shell, type `make BOARD=<chip> run_all` to build and load firmware into Atmosic EVK.

Shell and debug terminals after the `make BOARD=<chip> run_all` command is successfully executed.

```
@00000036 SDK Version: x.y.z
@0000008c APP Version: 0.0.0.9
6x6 EXT_FLASH: 4e 56 44 53 06 06 27 00 00 00 00 01 02 00 00 00 ...
@000001de boot_status = 1000000
@00000239 Cold boot
@00000274 [   BLE_adv][D]: user_main() done
@00000344 [   BLE_adv][D]: ble_adv_init: NVDS tag for adv timeout param not found. Using default
@000004f9 [   atm_adv][D]: Adv dur 0(unit:10ms) max_adv_evt 0 (timeout 0ms)
@00000608 [   BLE_adv][D]: adv_state = 2
@000006b6 [   BLE_adv][D]: adv_state = 4
@00000758 [   BLE_adv][D]: adv_state = 6
@000007e8 Optimize HW Scan: Active=1
@0000086a [   atm_adv][D]: Adv0: ON
@000008eb [   BLE_adv][D]: adv_state = 9
```

Detailed documentation for Beacon and other examples in SDK is available in platform/atm2/ATM2xx-x1x/examples/<example_name>/README file.

6.2 ATM33/e Device

At MSYS2 shell, go to the BLE_adv example folder to compile the project using the following command.

6.2.1 Board Configuration

The examples that depend on the pinmux driver directly or indirectly (via the GPIO, I2C, SPI, keyboard, PDM, GADC drivers) will need to specify the board for which the example is being built.

For ATM33xx devices, the EVB with specific SoC boards can be specified as:

- ATMEVK_3330_QN (for ATMEVK-3330-QN EVB)
- ATMEVK_3330e_QN (for ATMEVK-3330e-QN EVB)
- ATMEVK_3325_QK (for ATMEVK-3325-QK EVB)
- ATMEVK_3325_LQK (for ATMEVK-3325-LQK EVB)

6.2.2 Beacon

Beacon configuration: EddyStone, Non Connectable, Scannable, 100 ms Interval. To build and load firmware into the Atmosic EVK, follow below:

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```
$ cd platform/atm33/ATM33xx-5/examples/BLE_adv/
$ make BOARD=<chip> run_all
```

***Note**: Reminder to install COIN battery when VBATT = COIN OR configure EVB to run on VBATT = 3 V. Refer to the EVK User Guides for specific EVB for correct jumper settings.*

7. Building Energy Harvesting Beacon

7.1 ATM3 Devices

At MSYS2 shell, go to the BLE_harv_adv example folder to compile the project through the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 1s Interval.

```
$ cd platform/atm3/ATM32xx-x1x/examples/BLE_harv_adv/
$ make BOARD=<chip> DEBUG= run_all
```

7.2 ATM33e Devices

At MSYS2 shell, go to the BLE_harv_adv example folder to compile the project using the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 1s Interval.

```
$ cd platform/atm33/ATM33xx-5/examples/BLE_harv_adv/
$ make BOARD=<chip> DEBUG=  run_all
```

8. Uninstall Atmosic SDK

Follow the steps to uninstall Atmosic SDK. Also see Figure 23.

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1) Click the `Start Menu` and `Settings`.
2) In the `Settings` page, click the `Apps` item.
3) Find the `Atmosic SDK (<version_number>)` item in the `Apps & features` page.
4) Click the `Uninstall` button to start the Atmosic SDK uninstallation process.
5) The Atmosic RDI Interface will be removed during the uninstallation process. Please remove the Atmosic EVB before reinstalling the Atmosic SDK again.

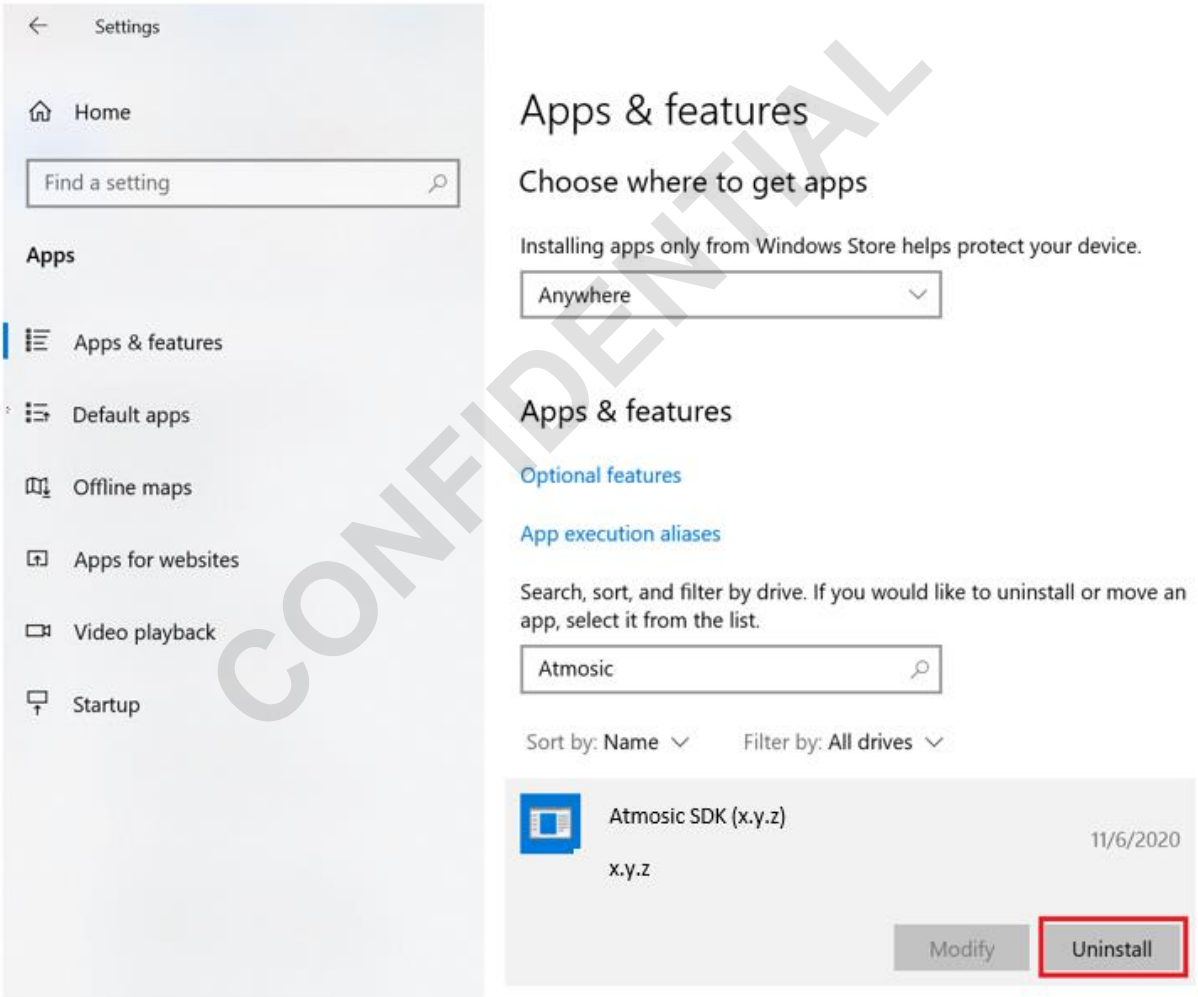


Figure 23 - Uninstall SDK

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Reference Documents

Title	Document Number
ATM2/ATM3 EVK Power Consumption Evaluation User Guide	ATM2_ATM3-UGPCE
ATM33e Series Energy Harvesting EVK Quick Start Guide	ATM33e-QSGEHV
ATM33/e Series Evaluation Kit User Guide	ATM33_e-UGEVK
ATM33/e Series EVK Power Consumption Evaluation User Guide	ATM33_e-UGPCE
EVK User's Guide for ATMx221	ATMx221-UG
EVK User's Guide for ATMx301/ATMx202	ATMx201-UG
IAR Workbench User Guide	ATM-UGIAR
Keil MDK Configuration Guide	ATM-CGKMDK
SEGGER Embedded Studio User Guide	ATM-UGSEG
IDE Auxiliary Flash Programming Tool User Guide	ATM-UGIDEFPT

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Revision History

Date	Version	Description
March 3, 2023	0.71	Added support of EVK for ATM3325 with embedded flash: Table 2 - Applicable ATM33/e EVKs, Board Configuration, Board Configuration. Updated Figure 1 - ATM33/e Evaluation Kit, Figure 2 - ATM33/e Evaluation Board, Figure 3 - ATM33/e Evaluation Board with 2.4 GHz Whip Antenna, Figure 22 - Atmosic Product Code on the EVB.
December 16, 2022	0.70	Added Linux and macOS Installation, Platform Tools & Utilities, Make Options and Commands sections. Renamed to SDK User Guide.
November 9, 2022	0.61	Added Remove RDI Interface section. Updated ATM33/e EVK, Building Beacon Application Using Atmosic SDK and Building Energy Harvesting Beacon sections.
May 13, 2022	0.60	Initial version created for SDK 5.1.0 release.

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